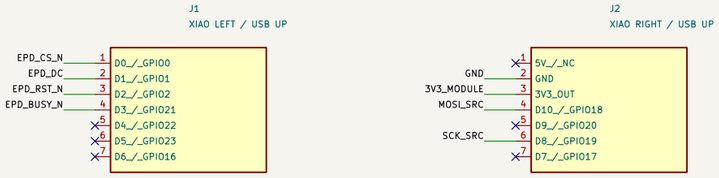
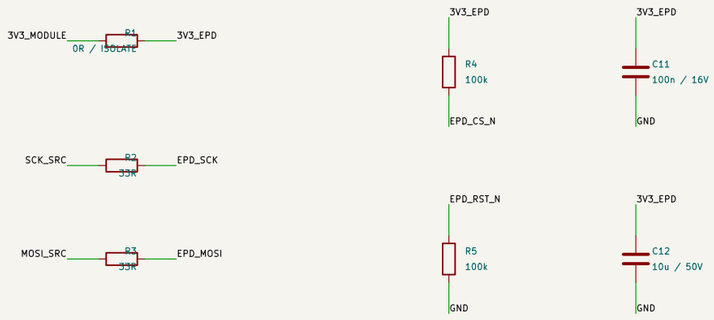
1 / SOCKETED XIAO – USB FACES TOP EDGE



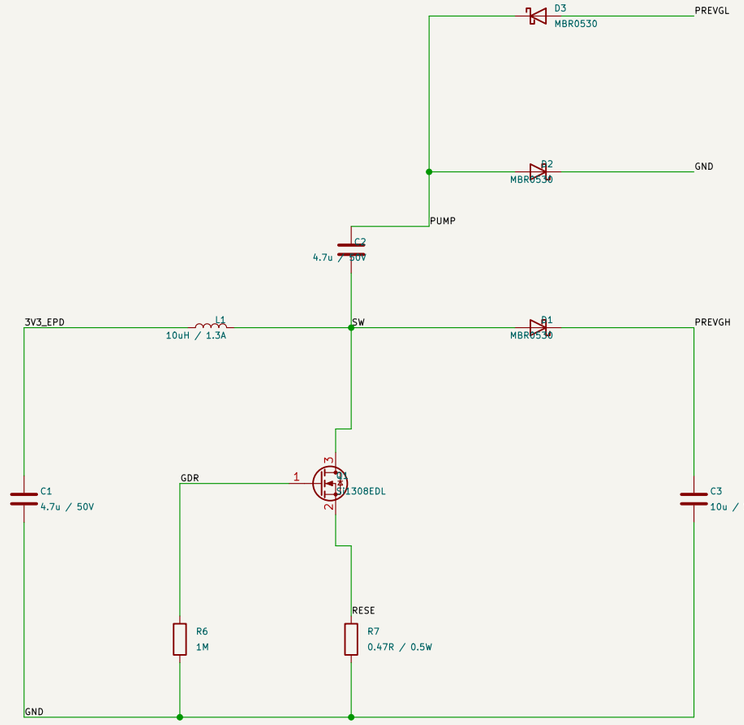
J1/J2 are physical female sockets. Pin 1 is nearest USB.
J2 pins are numbered top–to–bottom, not module DIP order.
15.24 mm row spacing; 2.54 mm pitch. No carrier 5V connection.



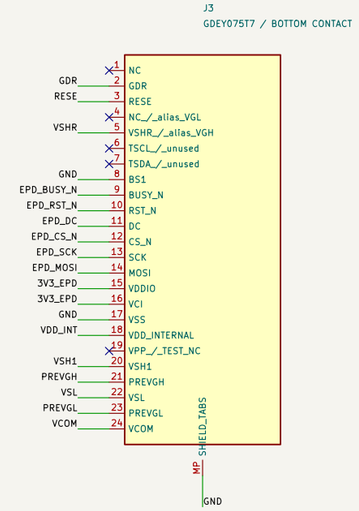
R1: remove to isolate / measure display supply.
Hold RESET low and CS high before SPI starts.
GPIO19=SCK, GPIO18=MOSI; start at 1 MHz.
Power off Wi–Fi before refresh.

2 / PANEL–CONTROLLED BOOST + CHARGE PUMP

Boost controlled by UC8179 through GDR / RESE.
Q1: 1=gate, 2=source, 3=drain. D1–D3: 1=cathode.
Kelvin pickup on RESE: short SW loop; 50V bias MLCCs.
Reference: Good Display GDEY075T7 rev1.0 p40.



3 / 24–PIN DISPLAY CONNECTOR



Pin 4 is NC. Pin 5 VSHR is NOT pin 21 PREVGH.
Pin 18 VDD_INTERNAL receives only C6, never external 3V3.
TSCL / TSDA / VPP unused per reference. BS1=GND.
Check flex orientation before inserting the panel.

4 / PANEL RAIL CAPACITORS

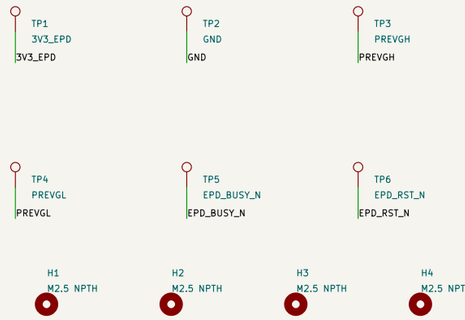


All rail capacitors non–polar MLCC. Bias–dependent capacitance must be checked against the exact panel lot.
No high–voltage rail is an external supply input. Fit panel and module only after inspection / resistance checks.

Four M2.5 non–plated holes. USB supply 5V / >=1A.
SMT top side only; sockets hand–soldered after reflow.



5 / PROBE POINTS



DESIGN REVISION A – PROTOTYPE NOT YET TESTED
GDEY075T7 / UC8179 | USB–powered socketed XIAO ESP32C6
Hector | Portfolio Engineering

Sheet: /
File: EPD_C6_RevA.kicad_sch

Title: ESP32–C6 E–paper Calendar Carrier

Size: A3	Date: 2026–10–20	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1